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Behavioural and nociceptive response in male and female spiny mice (*Acomys cahirinus*) upon exposure to snake odour

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Abstract

Predator cues (both mammalian odour or avian vocalizations) are known to elicit fear-associated responses in rodents, including analgesia. In previous studies it was reported that spiny mice fail to show fear responses when presented with the calls of an owl. In order to test the hypothesis that this species (living in semiarid and rocky areas) may react to stimuli coming from reptilian predators, 40 sexually mature spiny mice (20 males, 20 females) were individually exposed to a small cylinder containing either fresh sawdust or snake odour. Behavioural changes (5 min before and 15 min after odour exposure) as well as the subsequent performance in a hot-plate test ($50 \pm 0.5^\circ\text{C}$) were assessed. Results indicate that exposure to the odour of a sympatric terrestrial predator affected both behavioural and physiological responses of spiny mice. Upon exposure to snake odour both sexes showed significant changes in the patterns of inactivity, sniffing, grooming, sniffing the stimulus object (SO), withdraw reaction and in the frequency of somersaults. However, males increased the frequency of rearing, sniffing the SO, decreasing grooming more than females. No analgesic effect of odour exposure emerged; however, males showed significantly shorter latencies and higher frequencies of hindpaw licking compared to females. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Analgesia; Fear; Predator stimulus; Risk assessment; Sex differences; Spiny mouse

1. Introduction

The threat of predation is an ecologically relevant stimulus eliciting a variety of behavioural and physiological responses. These include risk

assessment, decision making, defensive behaviours and fear reactions, as well as opioid mediated analgesia (e.g. Kavaliers, 1988; Jedrzejewsky and Jedrzejewsky, 1990; Lima and Dill, 1990; Hendrie and Neill, 1991; Dell'Omo and Alleva, 1993).

In rodents endogenous pain-inhibitory mechanisms depend on both quantity and quality of the impinging stimuli (Bolles and Fanselow, 1982;

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Amit and Galina, 1986; Fanselow and Sigmundi, 1986). Physical stressors, such as enforced swim and noxious thermal or electric stimuli, produce marked and long-lasting analgesia (Grau et al., 1981; Fanselow, 1984; Disalver et al., 1990). Stressors of social origin also influence nociceptive sensitivity since, for example, odours released by stressed rats produce opioid analgesia in unstressed conspecifics (Rodgers and Hendrie, 1983; Fanselow, 1985). Endogenous opiate systems and benzodiazepine-sensitive neural mechanisms are reportedly involved in these behavioural regulations (Teskey et al., 1984; Morgan et al., 1987; Teskey and Kavaliers, 1988).

Species specific differences in reactivity to odour cues emanating from natural predators have been reported (Hendrie et al., 1996). Laboratory albino rats possess a hereditary stress response to predator odours, such as the red fox (*Vulpes vulpes*), as measured by behavioural (Vernet-Maury et al., 1968), electrophysiological (Cattarelli, 1977) and biochemical tests (Vernet-Maury, 1970). Moreover, exposure for 10 min to cat cues significantly reduces reactivity to a skin-irritating formalin injection in rats (Lester and Fanselow, 1985). Non-visual 30-s exposure to a short-tailed weasel (*Mustela erminea*) has been shown to induce benzodiazepine-sensitive analgesia in white-footed mice when subjected to the hot-plate test, while longer exposure (15 min) elicits long-lasting, naloxone-reversible analgesia in these animals. Studies using inbred DBA/2 mice indicated that acoustic stimuli, represented by calls emitted by murine predators such as the tawny owl (*Strix aluco*) and the barn owl (*Tyto alba*), inhibit consummatory behaviour, activate defensive mechanisms and elicit tail-flick analgesia involving both opiate and benzodiazepine-sensitive systems, the latter even upon very short exposure (Hendrie, 1991; Hendrie and Neill, 1991).

Sex differences in predator-induced responses have also been reported: male and female laboratory mice and rats differ in the levels of activity and analgesia displayed after exposure to a cat (Blanchard et al., 1991, 1995), while male meadow voles (*Microtus pennsylvanicus*) decrease spontaneous activity upon exposure to fox odour (Per-

rot-Sinal et al., 1996). A consistent difference in the predator stress response has been recently found in rats (Klein et al., 1994; Sheperd et al., 1997), females showing more defensive behavioural patterns than males upon both direct and indirect exposure to a cat. As for pain sensitivity, it has been well established that female rats have a lower threshold than males (assessments by electric shock sensitivities), a likely explanation being that gonadal hormones contribute to the difference by influencing body weight (Gandelman, 1983). Also laboratory CD-1 female mice show a higher sensitivity to thermal stimulation than males (Loggi et al., 1996). In meadow voles significant analgesic responses to a snake predator have been found in young, but not in adults, females displaying greater sensitive non-opioid analgesia than males (Saksida et al., 1993).

Few data on the effects of predator cues other than mammalian odours or avian vocalizations are available (Hendrie et al., 1996, but see Hendrie et al., 1998). Experiments conducted in laboratory mice using olfactory cues produced by *Elaphe obsoleta* showed behavioural changes (e.g. lowered food intake and higher defaecation rate), but no analgesic effect (Weldon et al., 1987; Dell'Omo and Alleva, 1993).

The spiny mouse (*Acomys cahirinus*) is a small rodent originating in semiarid areas of the Near and Middle East (Arnold and Burton, 1978; Delany and Happold, 1979). It feeds, also during daytime, on a mixed diet including snails, insects and seeds, but predation on bird eggs has been recently reported (Leader and Yom-Tov 1998). Little is known about social organization and reproductive habits (Delany and Happold, 1979; reviewed in Brunjes, 1990): spiny mice have been reported to live in groups, adopting a colonial lifestyle without nests, dominant females nursing all pups and excluding other mothers. Males are also involved in parental care, pups spending much more time with the father than with the mother (Makin and Porter, 1984). Adult females are more aggressive than males (Porter, 1976). As for the mating system no field data are available, although due to the colonial lifestyle polygyny can be hypothesized.

Both *Acomys dimidiatus* and *Acomys cahirinus* have been shown not to react to owl calls displaying enigmatic behaviour in apparently ignoring them (Hendrie et al., 1996, 1998). While emphasizing the need for further investigation, the same authors presented three possible explanations for the lack of reaction in this species: (1) the spines in the dorsal surface may dissuade predators from attacking them; (2) stimuli indicating the presence of an owl are irrelevant, since in natural settings the species would be rarely exposed to predatory attack from the air; (3) the animal may be reacting, but this was not measurable by behavioural observation.

The aim of the study was to test hypothesis (2) providing an assessment of the short term behavioural and nociceptive response in the spiny mouse (both sexes) to odour cues emanating from a reptilian species, the sand boa, *Eryx jaculus*. No data on snake predation on spiny mice are actually available; however, sand boas are known to eat small rodents and lizards which they catch by lying in wait nearly buried in the dirt or sand until a potential meal walks by, also searching out rodents to dine on the young (<http://www.tbnet.com/~gknaack/sbcs.html>). Moreover, this ophid shares the same habitat and geographical area of the spiny mouse (Arnold and Burton 1978). Therefore, a constant prey–predator interaction over evolutionary time may be assumed and behavioural changes including fear responses and risk assessment as well as stress-induced analgesia upon odour exposure are expected.

2. Materials and methods

2.1. Animals

The subjects of this study were 20 sexually mature male (30–45 g, mean $\pm$ S.D. = 37.4 ± 4.4) and 20 sexually mature female (25–40 g mean $\pm$ S.D. = 34.1 ± 3.9) spiny mice bred in the ENEA (AMB. PRO. TOSS., ENEA-Casaccia, Rome) laboratories from a colony established in 1987 and experimentally naive. At the time of testing they were 6–7 months old and were housed in

unisex groups of five to six individuals in $25 \times 25 \times 13$ cm Plexiglas boxes with pine sawdust as bedding material and maintained in a $4 \times 4 \times 3$ m air conditioned room ($T = 25 \pm 1^\circ\text{C}$; humidity $\sim 60\%$) with a natural 12:12 h light/dark cycle (lights off from 19:00 to 07:00 h). Pellet food and water were available ad libitum.

2.2. Procedure

Animals were individually isolated 15 days before the testing. All experimental manipulations were conducted between 10:00 and 14:00 h in a quiet laboratory room, isolated from the animal facility and maintained under similar temperature, light, and humidity condition.

Each subject was individually transferred in the experimental room 20 min before the test. A small black perforated plastic cylinder 5 cm long $\times$ 3 cm diameter (a new one used for each subject), the stimulus object (SO), was filled of faecal boluses, dry skin, and 10 g of ground material coming from the terrarium of a male sand boa of the Rome zoo. Faecal boluses were previously kept at -20°C and taken out 3 h prior to the experiment. The SO was introduced in one of the right corners of the home cage for 15 min. Control animals (ten males, ten females) were exposed in their home cages to a cylinder filled of 20 g of fresh sawdust.

After stimulus exposure, each animal was rapidly (about 10 s) introduced for 60 s in a hot-plate apparatus (Socrel Hot-Plate model DS 37: Ugo Basile, Italy) placed 50 cm away from the home cage and set at $50 \pm 0.5^\circ\text{C}$, in order to measure its pain sensitivity; the hot-plate was washed with clean water and dried after each single test. The temperature of 50°C , commonly used in mice and rats, was also chosen on the basis of studies carried out in species other than laboratory rats and mice, such as voles and deer mice (e.g. Saksida et al., 1993; Kavaliers et al., 1998), in which a response was elicited.

Each animal was videotaped with a JVC KY-17B 3-CCD color video-camera placed at a distance of 3 m. The behaviour of each animal (frequency and/or duration) prior to (5 min), during the exposure (15 min) and during the hot-plate test (60 s) was scored blind by a trained

operator using the software Observer 2.0 (Noldus, 1991) and according to the following ethogram partially derived from that used in the study of Birke and Sadler (1986): inactivity, locomotion (time spent moving), number of contacts with the SO, withdraw (rapid drawing back after approaching the SO), latency to the first and the second sniffing of the SO, sniffing, sniffing the object, feeding, grooming (mouth or paws on body), somersault, bar-holding, digging (moving the sawdust on the floor with the front paws), rearing (body in vertical position), wall rearing (front paws on the wall).

Pain sensitivity was assessed during the hot-plate test by measuring the latency to lick hind paw. Frequency and latency of wall rearing, rearing, jumping, forelimb licking and defecation during the hot-plate test were also recorded. The measurement of the latency of the response to a heated surface is a standard and ethically correct method of recording pain sensitivity, since it has been shown to have no adverse effects on foot tissue (Hunskar et al., 1986).

All mice were used only once and after the test they were immediately transferred in another room, since it has been demonstrated that stressors of social origin could influence nociceptive sensitivity (Rodgers and Hendrie, 1983, Fanselow, 1985).

Data were analyzed by a mixed model of ANOVA followed by Tukey's HSD post hoc considering sex and experimental condition (odour/no odour) as between subject factors, and time (divided into four 5-min blocks) as within subjects factor. For the hot-plate test and the latency measures the factors considered in the analysis were sex and experimental condition. All hypothesis tests used $\alpha = 0.05$ as criterion for significance.

3. Results

3.1. Behavioural assessment

All the parameters considered, except for somersault, bar-holding and digging, changed with time. Exposure to snake odour produced differences for

inactivity, grooming, sniffing, latency to and sniffing the SO, and somersault. Significant interactions including the sex of animals were found for inactivity, rearing, wall rearing, grooming, sniffing and sniffing the SO.

Specifically, inactivity levels decreased with time, being higher in odour exposed mice [odour $F(1, 36) = 4.9$, $P < 0.03$; Fig. 1]. Males experiencing the odour were more inactive [sex $\times$ odour $F(1, 36) = 11.6$, $P < 0.009$; Fig. 2]. Conversely, grooming tended to increase with time, but was observed less in odour exposed mice [odour $F(1, 36) = 6.7$, $P < 0.01$; Fig. 1, Table 1], males displaying less this activity upon exposure [sex $\times$ odour $F(1, 36) = 3.9$, $P < 0.05$; Fig. 2]. Feeding activity was affected more by the introduction of the SO per se rather than the snake odour, decreasing after the first 5-min block (Table 1). Both frequency and duration of sniffing were markedly inhibited by the presence of the SO (Fig. 1, Table 1; $P < 0.01$ in post hocs), while odour exposure reduced duration even further [$F(1, 36) = 6.1$, $P < 0.01$], an

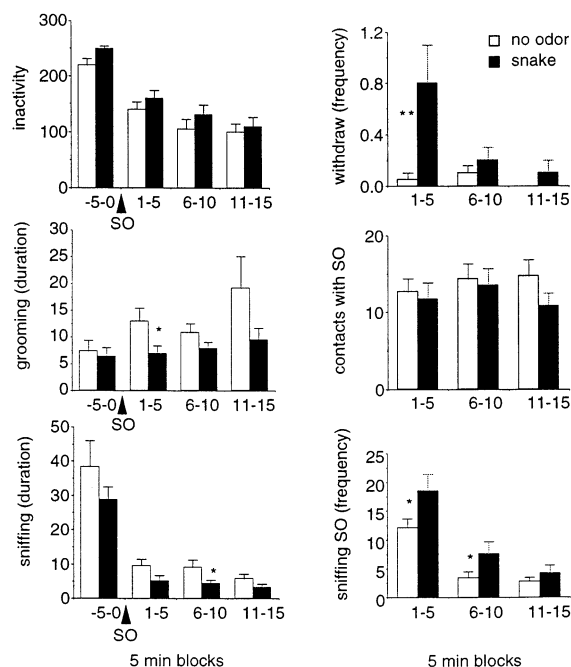


Fig. 1. Mean frequency or duration of various behavioural responses recorded in spiny mice prior and upon exposure to the stimulus object (SO; for details see Section 2). * $P < 0.05$; ** $P < 0.01$.

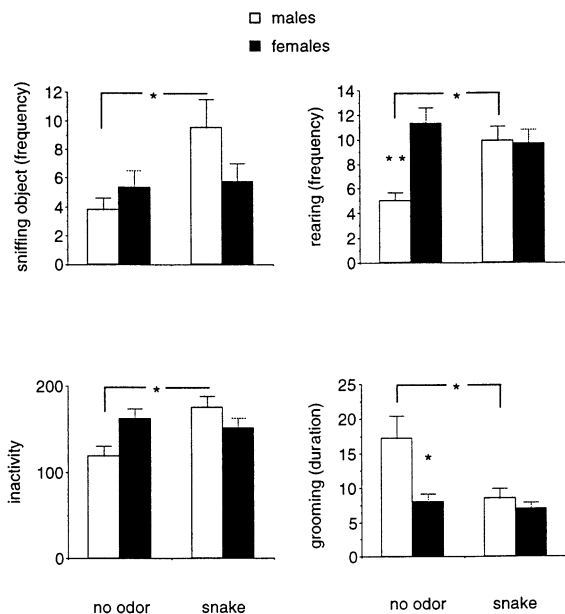


Fig. 2. Sex differences in behavioural responses recorded in spiny mice prior and upon exposure to the stimulus object. * $P < 0.05$; ** $P < 0.01$.

effect only seen in males [sex $\times$ odour $F(1, 36) = 7.7$, $P < 0.06$; not shown]. The significant interaction sex $\times$ odour $\times$ time [$F(3, 108) = 4.2$, $P < 0.006$; not shown] may indicate that females are more predisposed to habituate to the snake odour. Sniffing directed at the SO was more frequent in odour exposed animals [$F(1, 36) = 7.4$, $P < 0.007$; Fig. 1], particularly in males [sex $\times$ odour $F(1, 36) = 5.7$, $P < 0.02$; see Fig. 2 for post hocs], despite the overall number of contacts with the SO was significantly higher in females [sex $F(1, 36) = 15.4$, $P < 0.001$; not shown]; for this parameter no effect of odour was evident (Fig. 1). Digging was barely observed and both its frequency and duration did not vary with sex or experimental condition. However, its frequency seemed to decrease upon odour exposure [$F(1, 36) = 3.3$, $P < 0.07$; Table 1]. As for rearing, no main effect of odour, but an increase in frequency which failed to reach significance was evident [odour $F(1, 36) = 2.8$, $P < 0.09$; Table 1]; females consistently showed higher levels of rearing than males [sex $F(1, 36) = 10$, $P < 0.002$; not shown] and this picture changed upon exposure to snake

odour [sex $\times$ odour $F(1, 36) = 11.7$, $P < 0.001$; Fig. 2], males increasing frequency of rearing up to the levels of both control and exposed females. Frequency of wall rearing showed a similar figure. Reactions to snake odour were evident both as higher frequencies of somersaults [$F(1, 36) = 3.8$, $P < 0.05$; Table 1] or bar-holding, the latter score showing a trend towards significance [$F(1, 36) = 2.8$, $P < 0.08$, but $P < 0.05$ in post hocs; Table 1] and likely representing attempts to escape from the cage.

Finally, the most striking differences emerged in withdraw reaction to the SO in odour exposed animals as well as in the measures of latency to sniffing the SO after its introduction. Frequency of withdraw was about four-fold higher [odour $F(1, 36) = 6.5$, $P < 0.01$] than in controls, being characterized by an elevated responding in the first 5-min block after initial exposure followed by a rapid decrease to control levels [odour $\times$ time $F(3, 108) = 3.4$, $P < 0.02$; Fig. 1]. Latency to the first sniffing approach to the SO was shorter in odour exposed animals than in controls [$F(1, 36) = 8.5$, $P < 0.006$]; however these animals were slower than controls to approach the SO the second time [$F(1, 36) = 6.7$, $P < 0.01$], as it appears from Fig. 3. No sex difference was found for these parameters.

3.2. Hot-plate test

No abnormal responses (e.g. jumping) were observed during the test. No main effect of snake odour was found. On the whole, females showed higher latencies to lick hind paw [sex $F(1, 36) = 10.4$, $P < 0.003$], but considering the interaction of this variable with the sex, this difference was evident only in those mice which experienced snake odour [sex $\times$ odour $F(1, 36) = 3.1$, $P < 0.08$, $P < 0.01$ in post hoc; Fig. 4]. Conversely, frequency of licking was higher in males than in females [sex $F(1, 36) = 5.4$, $P < 0.02$], but this difference only emerged upon exposure to snake odour [odour $\times$ sex $F(1, 36) = 5.4$, $P < 0.02$; Fig. 4]. No significant effects were evident for the remaining parameters.

Table 1

Behavioural responses prior to (first 5-min block) and upon exposure to the stimulus object (SO) in both control and odour exposed spiny mice

Behavioural category	Group	First block (without SO)	Second block	Third block	Fourth block
Locomotion (duration)	No odour	14.6 ± 2.3	13.8 ± 1.9	8.6 ± 1.4	9.7 ± 2.3
	Snake	16.3 ± 1.7	11.4 ± 1.7	8.9 ± 1.3	8.1 ± 0.9
Sniffing (duration)	No odour	38.5 ± 7.4	9.6 ± 1.8	9.1 ± 2.2	5.8 ± 1.3
	Snake	28.9 ± 3.6	5.2 ± 1.4	4.5 ± 0.9	3.3 ± 0.9
Rearing (frequency)	No odour	14.2 ± 1.9	9.2 ± 1.3	5.7 ± 1.5	3.6 ± 0.8
	Snake	17.3 ± 1.9	10.4 ± 1.4	6.2 ± 1.1	5.2 ± 1.1
Wall rearing (frequency)	No odour	27.8 ± 2.1	14.4 ± 1.9	8.1 ± 1.6	6.5 ± 1.2
	Snake	30.9 ± 2.8	12.9 ± 2.2	12.1 ± 2.6	7.7 ± 1.5
Bar-holding (frequency)	No odour	0.2 ± 0.1	0.1 ± 0.1	0.05 ± 0.05	0.0
	Snake	0.3 ± 0.2	0.1 ± 0.1	1.7 ± 1.4	0.7 ± 0.6
Somersault (frequency)	No odour	0.05 ± 0.05	0.1 ± 0.06	0.6 ± 0.5	0.2 ± 0.1
	Snake	0.4 ± 0.4	1.8 ± 1.2	2.4 ± 1.9	1.5 ± 0.9
Feeding (frequency)	No odour	2.1 ± 0.6	0.7 ± 0.2	0.8 ± 0.6	0.7 ± 0.3
	Snake	1.6 ± 0.3	0.6 ± 0.2	0.8 ± 0.6	0.3 ± 0.1
Digging (frequency)	No odour	0.8 ± 0.4	1.1 ± 0.5	0.9 ± 0.4	0.6 ± 0.3
	Snake	0.6 ± 0.3	0.8 ± 0.4	0.2 ± 0.08	0.6 ± 0.3
Grooming (duration)	No odour	7.4 ± 2.0	13 ± 2.4	10.8 ± 1.7	19.2 ± 5.9
	Snake	6.4 ± 1.8	7 ± 1.6	7.9 ± 1.3	9.5 ± 2.2
Inactivity (duration)	No odour	219.7 ± 11.6	139.8 ± 14.1	105 ± 17	99.7 ± 14.6
	Snake	249.7 ± 4.8	160.4 ± 13.7	131.3 ± 17	109.6 ± 16.5

4. Discussion

The purpose of this study was to examine the behavioural and physiological responses to a reptilian predator stimulus (including sex differences) in a species previously shown not to react to an avian predator cue (Hendrie et al., 1998).

A possible confound could arise because the faeces we used could have been less effective than fresh faeces. Obviously dry faecal material smells less than fresh material. In carnivores (mammalia) the component of excreta which are believed to be implicated in triggering antipredator behavioural or physiological responses are fatty acids, which are mostly volatile compounds. The faeces we used were emitted by a snake housed in a terrarium of the zoo that had eaten the previous day. They were collected and immediately frozen at -20°C . They could not be considered fresh given that we did not collect them immediately after defecation and they were defrosted 3 h prior to the experiment. However, defaecation rate in snakes is known to be very low (days or even weeks) and the probability that in the real world a

spiny mouse would incur very fresh excreta is likely to be low.

Results clearly show that snake odour was rapidly detected in the initial 5 min of exposure by both sexes, but the intensity and the quality of the responses varied throughout the exposure time and, interestingly, with sex. Exposing the animals to the snake odour produced a marked behavioural change consisting of a six-fold increase

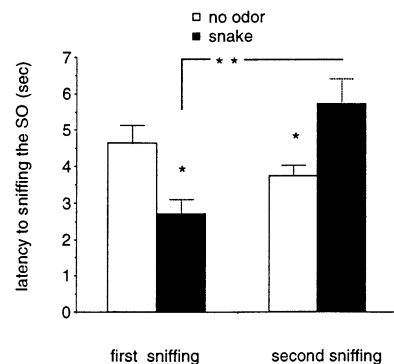


Fig. 3. First and second latency to sniffing the stimulus object (SO). * $P < 0.05$; ** $P < 0.01$.

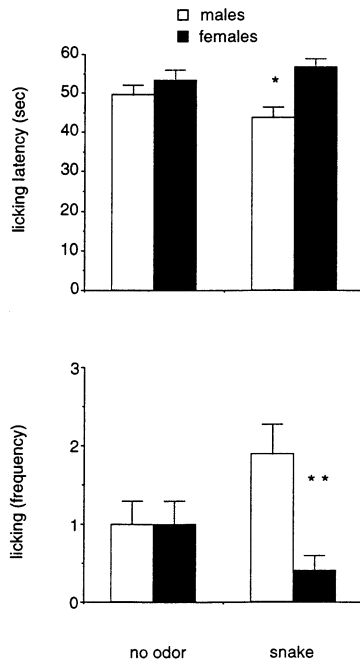


Fig. 4. Frequency and latency to lick the hind paw in both control and odour exposed spiny mice during the hot-plate test (for details see Section 2). * $P < 0.05$; ** $P < 0.01$.

in withdraw, and an increase in frequency of sniffing the SO which persisted also in the second 5-min block. Moreover, exploratory and risk assessment behaviours in response to novelty were clearly elicited by the introduction of the SO per se. However, the first two approaches to the SO (latencies to sniffing) showed remarkable differences between control and odour exposed mice: the latter showed a shorter latency to the first approach of the SO, likely due to the immediate need for investigation, gaining information and assessing possible risks (exploratory behaviour), while thereafter, time before the second sniffing approach was longer than in controls, indicating that after detection of an actual threat a fear response was evoked.

An analgesic effect was previously reported for mice and rats upon exposure to several kinds of stressors, while in the present study we only recorded the emergence of a sex difference in nociception. In fact upon snake odour exposure, males showed lower pain thresholds than females.

These data seem to agree with previous studies on sex differences in pain sensitivity (see Section 1), and may indicate that females are more stressed upon odour exposure (which occurred immediately prior to the hot plate test), leading to the hypothesis that an activation of opioid and/or non-opioid pain inhibitory mechanisms occurred (Hendrie, 1991; Hendrie and Neill, 1991). Furthermore, our findings are pertinent with those found on meadow voles, where significant analgesic response, including a sex difference (young females with greater non-opioid sensitive analgesia; Saksida et al., 1993) have been found. Data on nociception agree with literature reports indicating that female rodents show higher heart rates than males following open field testing and during handling (rats), as well as freeze, flee and defecate more after a change in stimulation (mice). According to a more recent report, female rats are more stressed than males, being more inactive and engaging less time in grooming upon exposure to cat urine (Klein et al., 1994). We cannot exclude that the temperature level we used for the hot-plate test ($50 \pm 1^\circ\text{C}$), according to the current regulations, could have masked a more pronounced analgesic effect, considering also that the *Acomys* genus has also diurnal habits and it is adapted for an arid lifestyle, living in rocky areas of desert regions (Delany and Happold, 1979). It is known that *Acomys russatus* is physiologically adapted to heat stress (Shkolnik and Borut, 1969) and foot skin could be more resistant to heated surfaces than in other rodent species. Nonetheless, predator-induced analgesia has been successfully measured with the temperature we used (e.g. Saksida et al., 1993; see Section 2).

Spiny mice have been observed to show a great readiness to explore novel objects under laboratory conditions and to exhibit patterns of exploration which differ respect to the usually much more cautious laboratory mice (Birke et al., 1981). As regards sex differences, males are reported to be more cautious than females, the latter performing more rearing than males during exploration of a novel environment (Birke and Sadler, 1986). In the present study, while the levels of rearing after the introduction of the SO confirmed the above-mentioned sex difference in control subjects, upon

odour exposure males were ready to explore at least like females, as shown by the increased frequency of rearing. Such a discrepancy could be explained taking into account that, compared to a novel object, the odour of a predator represents a much evocative stimulus, likely fixed in the 'genetic memory' of the species. Other behavioural responses were more affected in males than in females. Specifically, males increased their levels of sniffing, wall rearing, inactivity and decreased time spent in grooming, while females did not change their behaviour significantly. Thus, although no sex difference was recorded for the latency to sniffing the SO, the data seem to suggest that snake odour affected male behaviour more than females. The increase in rearing, wall rearing and sniffing, characterizing the behaviour of male subjects exposed to snake odour, can be interpreted as attempts to patrol the territory, a typical male rodent activity. Spiny mice show a colonial lifestyle where sex roles are likely to be different (see Section 1). Males seem to adopt an adaptive behavioural pattern which fulfils the need to explore and patrol a territory, while females do not change their activities significantly, according to the need to be less visible, a behaviour more convenient for individuals whose primary need is nest and offspring defence at least during certain phases of parental care.

Behavioural data on sex differences are not in total agreement with those obtained in laboratory rats and mice (Archer, 1975) in which females exposed to fear-evoking situations are reported to show more intense initial responses associated with higher level of activity than males. In particular, they are described to ambulate, sniff and rear more than males in a novel situation (mice to a lesser extent). However, the fact that males, the more active individuals in this likely polygynous species, show greater responses is consistent with predator responses of other diurnal polygynous species such as meadow voles (Perrot-Sinal et al., 1996).

On the whole, the data indicate that the threat of predation represented by exposure to the odour of a sympatric terrestrial predator affected both behavioural and physiological responses of spiny mice. In particular, males reacted with explo-

ration and risk assessment behaviours, while females showed a slightly pronounced stress-induced analgesia. According to these results, the hypothesis that owl stimuli were irrelevant because the *Acomys* gene pool has not co-evolved with this kind of predator seems feasible. As a consequence, of the three hypotheses proposed to explain the ineffectiveness of owl vocalization in inducing a response in spiny mice (see Section 1), the second appears to be the most reasonable. Conclusions have, however, to be restricted due to some shortcomings of the experimental design used in this study. In particular, a novel non-predator odour should be used as a further control for distinguishing novelty from potential predator threat. Finally, in view of the species-specific differences in reactions to the threat of predation already observed (Hendrie et al., 1996), it is worth emphasizing the need for further studies on rodent species, whose ecology and behaviour is sufficiently known, other than laboratory mice and rats. A more naturalistic approach aimed at detecting both intraspecific (sex, age, interindividual differences) and interspecific variation in the way animal species cope with environmental challenges is also recommended (Alleva et al., 1995, 1996).

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